



TeamUp
Consultants

CYBERSECURITY SKILL CHECKLIST



**A Practical Guide to
Identifying, Assessing,
and Developing
Cybersecurity Skills**



COMPLETE SKILL FRAMEWORK
Covering 20+ domains and
70+ key cybersecurity skills.



BEGINNER TO EXPERT LEVELS
Understand skill progression
at every level.



ASSESSMENT & EVIDENCE
Learn how to evaluate skills
with real examples and methods.



SKILL GAP ANALYSIS
Identify gaps and create
targeted development plans.



ROLE-SPECIFIC CHECKLISTS
Detailed skills for 20+ cybersecurity
roles and career paths.



PRACTICAL & ACTIONABLE
Real-world scenarios, case studies,
and master checklists.



PROTECT
Systems
Data
People

DETECT
Threats
Risks
Vulnerabilities

RESPOND
Incidents
Analysis
Recovery



**BUILD SKILLS.
ASSESS CAPABILITIES.
SECURE THE FUTURE.**

**YOUR
LOGO
HERE**

**TEAMUP
CONSULTANTS**

Practical Cybersecurity
Hiring Resources



KNOWLEDGE
Understand
Concepts



PRACTICE
Apply in
Practice



ASSESS
Measure
Proficiency



IMPROVE
Grow
Continuously

Introduction & Conceptual Framework

Cybersecurity is one of the most critical, dynamic, and complex disciplines in modern technology. As digital infrastructure expands, threat actors grow more sophisticated, and regulatory mandates tighten, organizations face an unprecedented shortage of qualified professionals. However, hiring, assessing, and developing talent remains notoriously difficult because "cybersecurity" is not a single monolith—it is a vast ecosystem of interconnected technical domains, operational workflows, governance frameworks, and behavioral competencies.

This handbook - **Cybersecurity Skill Checklist** - is designed to solve this evaluation challenge. Developed by TeamUp Consultants, this guide provides a structured, rigorous, and practical framework for identifying, assessing, and cultivating cybersecurity skills across every major domain, seniority level, and organizational role.

Core Definitions: Knowledge vs. Skill vs. Experience vs. Competency

To evaluate talent accurately, we must distinguish between five foundational concepts that are often conflated:

- **Knowledge:** Understanding theoretical concepts, definitions, and principles (e.g., knowing what the CIA triad stands for or how TLS encryption works).
- **Skill:** The practical ability to apply knowledge to perform a specific task successfully (e.g., writing a YARA rule, analyzing a Wireshark packet capture, or configuring an AWS Security Group).
- **Experience:** The accumulated exposure and contextual duration working through real-world scenarios, challenges, and operational environments.
- **Competency:** The integrated capability to combine knowledge, skills, and judgment to solve complex, ambiguous problems independently.
- **Capability:** An organizational or individual measure of readiness to achieve desired security outcomes against real threats.

Organizations must evaluate cybersecurity skills systematically because random hiring leads to capability gaps, operational blindness during security incidents, and failed audits. Whether you are a fresher looking to break into the industry, an experienced engineer targeting leadership, a

recruiter screening resumes, or a CISO building a world-class security operations center, this book provides the exact benchmarks needed.

Who Should Use This Checklist

Different stakeholders can leverage this checklist in targeted ways to maximize its utility:

- **Students & Freshers:** Use the skill checklists to build a targeted learning roadmap, set up home labs, build portfolios, and align self-study with industry demands.
- **Experienced Professionals:** Use the advanced matrices to identify skill gaps, prepare for senior promotions, transition into specialized roles (such as Cloud Security or Penetration Testing), and validate expertise.
- **Recruiters & HR Teams:** Use the role-specific checklists and interview frameworks to screen resumes effectively, formulate initial screening questions, and separate genuine technical capability from buzzword stuffing.
- **Hiring Managers & Security Leaders:** Use the technical assessments, scenario-based evaluations, and skill-gap frameworks to conduct rigorous technical interviews and build balanced teams.

Key Principle: Cybersecurity requires a balance of technical depth, analytical problem-solving, ethical grounding, and clear communication. A brilliant engineer who cannot communicate risk to executive stakeholders creates significant organizational vulnerability.

Part 1 - Cybersecurity Skill Foundations

Before diving into advanced engineering, offensive security, or governance, every cybersecurity professional must master foundational concepts. These core principles govern all security architecture, defensive operations, and risk management.

The CIA Triad

The cornerstone of information security is the CIA Triad: Confidentiality, Integrity, and Availability. Every control, policy, and technical implementation maps back to protecting one or more of these three pillars.

- **Confidentiality:** Ensuring that sensitive data is accessed only by authorized individuals. Assessed through understanding encryption standards, access controls, and data classification.
- **Integrity:** Guaranteeing that data and systems remain accurate, complete, and uncompromised by unauthorized tampering. Assessed through hashing, digital signatures, and version control.
- **Availability:** Ensuring that systems, networks, and data are accessible to authorized users whenever required. Assessed through high availability, redundancy, and disaster recovery planning.

Access Control Principles

Controlling who accesses what is fundamental to stopping lateral movement during a breach. Professionals must understand and apply:

- **Least Privilege:** Users, programs, and systems should only have the minimum necessary permissions required to perform their authorized functions.
- **Defense in Depth:** Implementing multiple layers of security controls (perimeter defense, network segmentation, host hardening, application controls) so that if one layer fails, subsequent layers contain the threat.
- **Separation of Duties:** Dividing critical tasks among multiple people to prevent fraud, errors, or unauthorized unilateral actions.

Practical Application & Assessment

Scenario: An employee requests administrative access to the corporate financial database to install a printer driver. **Weak Response:** Granting admin access because the user asked politely. **Strong Response:** Denying the request based on least privilege; providing a temporary, limited execution context or having IT install the driver without elevating user privileges.

Part 2 - Network Security Skills

Network security forms the digital perimeter and internal highways of an enterprise. Professionals must understand how data traverses networks and how to inspect, secure, and monitor traffic.

Skill Area	Beginner Expectation	Intermediate Expectation	Advanced Expectation
TCP/IP & OSI Model	Understand the 7 layers of OSI and TCP 3-way handshake.	Analyze TCP flags (SYN, ACK, FIN, RST) in packet captures.	Diagnose complex network latency, asymmetric routing, and protocol anomalies.
DNS & DHCP	Explain DNS resolution and DHCP lease processes.	Detect DNS tunneling, cache poisoning, and rogue DHCP servers.	Architect secure internal DNS infrastructure with DNSSEC and secure logging.
Firewalls & Segmentation	Understand stateful packet filtering and basic rule creation.	Design secure VLAN architectures and write restrictive firewall ACLs.	Implement Next-Generation Firewalls (NGFW), micro-segmentation, and zero-trust policies.
Packet Analysis (Wireshark)	Capture traffic and filter by IP or port.	Extract files, identify cleartext credentials, and spot port scans.	Perform deep forensic protocol analysis during active data exfiltration incidents.

Common Network Security Mistakes: Relying solely on perimeter firewalls while leaving internal networks flat and unsegmented; failing to log east-west traffic; using default credentials on network management interfaces.

Part 3 - Operating System Security

Endpoints are the primary target for modern adversaries. Securing Windows and Linux operating systems requires rigorous configuration management, logging, and hardening.

Linux Security & Hardening

Linux powers the majority of cloud infrastructure and servers. Professionals must demonstrate deep competency across distributions:

- **File Permissions & Ownership:** Mastery of `chmod`, `chown`, `setuid`, `setgid`, and Access Control Lists (ACLs).
- **Process & Service Management:** Using `systemd`, checking listening ports with `ss` or `netstat`, and identifying rogue background daemons.
- **Logging & Auditing:** Configuring `auditd`, analyzing `/var/log/auth.log`, and forwarding logs to a central SIEM.
- **System Hardening:** Disabling unnecessary services, enforcing SSH key-based authentication, configuring SELinux/AppArmor in enforcing mode.

Windows Security & Active Directory

Enterprise environments rely heavily on Windows Active Directory (AD) and Group Policy Objects (GPOs):

- **Event Logging:** Enabling and analyzing Windows Security Event Logs (e.g., Event ID 4624 for successful logons, 4688 for process creation, 4720 for user account creation).
- **PowerShell Security:** Utilizing PowerShell logging (Script Block Logging, Transcription) and detecting malicious obfuscated execution.
- **Active Directory Hardening:** Mitigating common misconfigurations like Kerberoasting, AS-REP roasting, and excessive Domain Admin privileges.

Assessment Question: "A Linux web server has been compromised via an uploaded PHP shell. Walk me through the exact commands and log files you would inspect to determine how the attacker gained initial execution and what files were modified."

Part 4 - Security Monitoring & SOC Skills

The Security Operations Center (SOC) is the frontline of defense. Tiered analysts must triage alerts, analyze telemetry, and escalate true incidents.

SOC Analyst Skill Tiers

- **Junior SOC Analyst (Tier 1):** Proficient in alert triage, reviewing SIEM dashboards, distinguishing false positives from true positives, executing initial playbooks, and creating clear ticketing documentation.
- **Mid-Level SOC Analyst (Tier 2):** Capable of deep-dive alert analysis, endpoint forensics using EDR tools (CrowdStrike, Defender for Endpoint), threat hunting via IOC matching, and rapid incident escalation.
- **Senior SOC Analyst / Lead (Tier 3):** Expert in malware triage, advanced adversary tracking, custom detection engineering (writing Sigma rules or Splunk SPL queries), and leading incident containment.

SOC Function	What Good Looks Like	What Weak Looks Like
Alert Triage	Quickly correlates multiple log sources, verifies asset criticality, and assesses business impact.	Mindlessly closes tickets as "Benign Positive" without checking supporting endpoint telemetry.
Documentation	Writes precise timeline of events, affected systems, root cause, and remediation steps in the ticket.	Writes vague notes like "Investigated alert, seemed fine" with no supporting artifacts.

Part 5 - Incident Response Skills

When defenses fail, Incident Response (IR) ensures containment, eradication, and recovery while preserving evidentiary integrity.

The 6 Phases of Incident Response (NIST Framework)

1. **Preparation:** Establishing IR plans, playbooks, communication channels, and tooling before an incident occurs.
2. **Detection & Analysis:** Identifying anomalous activity, scoping the compromise, and determining severity.
3. **Containment:** Isolating affected hosts from the network (e.g., network disconnect or EDR host isolation) to stop lateral movement while preserving volatile memory.
4. **Eradication:** Removing malware artifacts, closing vulnerability vectors, and resetting compromised credentials.
5. **Recovery:** Restoring systems from verified clean backups, validating system integrity, and returning to normal operations.
6. **Lessons Learned:** Conducting post-incident reviews, updating playbooks, and improving detection engineering.

Incident Scenario Example

Situation: Ransomware encrypts file servers at 2:00 AM. **Candidate Action:** Immediately isolate infected servers via EDR, identify patient zero and initial access vector (compromised VPN credentials), preserve memory dumps, notify legal and executive stakeholders, and initiate offline backup recovery.

Part 6 - Threat Intelligence and Threat Analysis

Understanding who is attacking your organization, what motives they possess, and what Tactics, Techniques, and Procedures (TTPs) they employ allows organizations to move from reactive defense to proactive hunting.

- **MITRE ATT&CK Framework:** Mapping adversary behaviors from Initial Access to Exfiltration. Analysts must be able to describe how specific techniques (e.g., T1078 Valid Accounts or T1566 Phishing) operate and how to detect them.
- **Indicators of Compromise (IOCs) vs. Indicators of Attack (IOAs):** IOCs (hashes, IPs, domains) tell you what happened in the past; IOAs (behaviors, unexpected process spawn chains) catch active intrusions in real-time.

- **Threat Hunting:** Hypothesis-driven searching through enterprise datasets to detect advanced adversaries who bypass automated security controls.

Part 7 - Vulnerability Management

Vulnerability management is the systematic process of identifying, classifying, prioritizing, remediating, and mitigating software and hardware vulnerabilities.

Crucial Distinction: A vulnerability is *not* automatically equivalent to a risk. A critical vulnerability on an isolated, air-gapped test machine with no network connectivity carries near-zero operational risk, whereas a medium vulnerability on an internet-facing payment gateway with active exploit code available demands immediate patching.

Key Skills Required: Mastery of vulnerability scanners (Tenable Nessus, Qualys, OpenVAS), understanding CVSS (Common Vulnerability Scoring System) vector strings, managing false positives, coordinating patch rollouts with IT operations, and verifying remediation through validation scans.

Part 8 - Cloud Security Skills

As organizations migrate to AWS, Azure, and Google Cloud Platform (GCP), traditional perimeter security is replaced by identity-centric cloud security architectures.

Cloud Security Domain	Beginner	Intermediate	Advanced
Identity & Access Management (IAM)	Create users and assign basic managed policies.	Design least-privilege custom IAM policies, cross-account roles, and SCPs.	Architect enterprise cloud federation, SSO, and automated temporary credential generation.

Cloud Networking & Firewalls	Configure Virtual Private Clouds (VPCs), subnets, and Security Groups.	Implement Transit Gateways, Network Firewall inspections, and private endpoint routing.	Design multi-region zero-trust cloud network architectures with strict egress filtering.
Shared Responsibility Model	Explain the division of security <i>in</i> the cloud vs. <i>of</i> the cloud.	Apply cloud-native compliance controls and configure CSPM tooling.	Build automated governance guardrails that prevent misconfigurations before deployment.

Part 9 - Application Security Skills

Application Security (AppSec) ensures that software is designed, written, and tested to be secure against attacks.

- **OWASP Top 10:** Deep mastery of injection flaws, broken authentication, sensitive data exposure, security misconfigurations, and broken access control.
- **Static & Dynamic Testing:** Integrating SAST (Static Application Security Testing) and DAST (Dynamic Application Security Testing) into CI/CD pipelines.
- **Threat Modeling:** Using methodologies like STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege) during the design phase of software development.

Part 10 - Identity and Access Management (IAM)

"Identity is the new perimeter." IAM professionals secure the digital identities of workforce users, external partners, and machine workloads.

Core competencies include implementing Multi-Factor Authentication (MFA), Single Sign-On (SSO) via SAML/OIDC, Role-Based Access Control (RBAC), Attribute-Based Access Control

(ABAC), Privileged Access Management (PAM) vaulting, and automated Joiner-Mover-Leaver (JML) identity lifecycles.

Part 11 - Security Engineering & Architecture

Security Engineers build and maintain resilient technical systems. They translate security policies into automated technical controls, deployment pipelines, and protective appliances.

Key skills include infrastructure-as-code security (Terraform scanning), endpoint security agent deployment, SIEM log pipeline tuning, network security monitoring architecture, and resilient cryptography implementation.

Part 12 - Penetration Testing & Offensive Security

Offensive security professionals simulate real-world cyberattacks to uncover vulnerabilities before malicious actors exploit them.

- **Phases of Testing:** Reconnaissance, Scanning, Enumeration, Exploitation, Privilege Escalation, and Professional Reporting.
- **Ethical Boundaries:** Operating strictly within the rules of engagement, avoiding destructive actions, protecting client data, and adhering to responsible disclosure guidelines.

Part 13 - Digital Forensics -->

Digital Forensics and Incident Response (DFIR) specialists recover and analyze digital evidence from computers, mobile devices, and cloud environments while maintaining a strict chain of custody for legal proceedings.

Part 14 & 15 - Governance, Risk, Compliance & Frameworks

Governance, Risk, and Compliance (GRC) professionals align cybersecurity posture with business objectives and regulatory mandates.

- **Key Frameworks:** NIST Cybersecurity Framework (CSF), NIST SP 800-53, ISO/IEC 27001, CIS Critical Security Controls, SOC 2 Type II, PCI DSS, and GDPR.
- **Core Tasks:** Conducting risk assessments, maintaining risk registers, managing third-party vendor risk, conducting control testing, and gathering audit evidence.

Parts 16 to 20 - Advanced Technical Domains

Modern cybersecurity demands mastery across specialized technical verticals:

- **Cryptography & Data Protection:** Symmetric vs. asymmetric encryption, hashing algorithms (SHA-256), digital signatures, Public Key Infrastructure (PKI), and secure secrets management.

- **Automation & Scripting:** Writing Python, PowerShell, and Bash scripts to automate log processing, alert enrichment, and SOAR (Security Orchestration, Automation, and Response) playbooks.
- **DevSecOps:** Shifting security left by embedding automated SAST, SCA (Software Composition Analysis), container scanning, and security gates into CI/CD pipelines.
- **Security Architecture:** Designing Zero Trust architectures, network micro-segmentation, and resilient defense-in-depth topologies.
- **Risk Decision-Making:** Evaluating quantitative and qualitative risk, calculating likelihood and impact, and presenting actionable risk treatments (mitigate, transfer, avoid, accept) to business executives.

Parts 21 to 24 - Communication, Problem-Solving & Soft Skills

Technical brilliance alone is insufficient for career success in cybersecurity. Professionals must master human-centric skills:

- **Communication:** Translating complex technical vulnerabilities (e.g., remote code execution via deserialization) into clear business risk metrics for C-level executives and board members.
- **Problem-Solving:** Applying methodical root-cause analysis, hypothesis testing, and rigorous troubleshooting during high-pressure incidents.
- **Ethics & Mindset:** Maintaining uncompromising integrity, respecting confidentiality, practicing responsible disclosure, and understanding the ethical boundaries of offensive and defensive security.
- **Collaboration & Leadership:** Demonstrating ownership, mentoring junior staff, managing cross-functional conflicts, and exhibiting continuous learning agility.

Part 25 - Role-Specific Skill Checklists

Different roles demand distinct skill profiles. Below is an evaluation breakdown for key security positions:

Role Title	Must-Have Core Skills	Key Tools	Common Skill Gaps
SOC Analyst	Log analysis, SIEM navigation, incident triage, TCP/IP fundamentals.	Splunk, QRadar, CrowdStrike EDR, Wireshark.	Inability to explain underlying network protocols; poor ticketing notes.
Cloud Security Engineer	IAM policy design, cloud networking, CSPM, container security.	AWS IAM, Terraform, Prisma Cloud, Docker/ Kubernetes.	Treating cloud security like traditional on-prem datacenter security.
Penetration Tester	Web app testing, network enumeration, privilege escalation, reporting.	Burp Suite, Metasploit, Nmap, Cobalt Strike.	Relying exclusively on automated scanner output without manual verification.
GRC Analyst	Risk assessment, control mapping, audit preparation, policy writing.	Archer, OneTrust, Excel, Jira/ Confluence.	Lack of technical context when evaluating actual system controls.

Parts 26 & 27 - Freshers & Experienced Professionals

For Freshers: Priority should be placed on mastering Linux command line, TCP/IP networking fundamentals, basic Windows administration, CompTIA Security+ or similar baseline certifications, and building a GitHub portfolio or home lab documenting network packet captures and log analysis exercises.

For Experienced Professionals: Progression requires demonstrating measurable business impact (e.g., "Reduced mean time to detect by 40%"), architectural ownership, incident leadership, mentorship of junior analysts, and active participation in threat intelligence communities.

Parts 28 to 32 - Assessment, Evidence & Gap Analysis

Assessing skills requires a rigorous, repeatable framework moving from theory to practical demonstration.

The 5 Levels of Skill Mastery

- **Level 1 (Understand):** "I know it." Familiar with definitions and theory.
- **Level 2 (Apply):** "I have used it." Hands-on experience in labs or guided tasks.
- **Level 3 (Troubleshoot):** "I can troubleshoot it." Able to diagnose and fix errors independently.
- **Level 4 (Design):** "I can design it." Capable of architecting solutions from scratch.
- **Level 5 (Lead/Teach):** "I can lead and teach it." Expert mastery; can mentor others and set strategy.

Skill Gap Formula: For every candidate or employee, map: [Required Skill] → [Current Level] vs. [Target Level] → [Calculated Gap] → [Business Impact] → [Actionable Development Plan].

Parts 33 to 38 - Practical Assessment & Interview Frameworks

Recruiters and hiring managers must use structured evaluation rubrics to eliminate bias and accurately gauge candidate capability.

Interview Question	What It Tests	Strong Answer Indicators	Weak Answer Indicators
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"Walk me through what happens when you type a URL into a browser."	Network fundamentals, DNS, TLS, HTTP.	Explains DNS lookup, TCP handshake, TLS negotiation, HTTP request/response clearly.	Vague hand-waving; cannot explain certificates or port numbers.
"How would you investigate a suspicious PowerShell process execution alert?"	Endpoint forensics, logging, analytical thinking.	Mentions checking command-line arguments, script block logs, parent-child process tree, network connections.	Says "I'd reimage the machine immediately" without investigation.

Red Flags to Watch For: Tool-name dropping without understanding underlying mechanisms; claiming expert knowledge on certifications without practical labs; inability to explain past projects or incident investigations in detail.

Parts 39 to 44 - Development Plans, Case Studies & Master Checklist

Developing cybersecurity capability is an ongoing journey. Structured 30/60/90-day plans ensure continuous growth.

Master Skill Evaluation Checklist

Use the following checklist to evaluate core domains across your organization or personal career:

- ☐ **Foundational Security:** CIA Triad, Least Privilege, Defense in Depth implemented and understood.
- ☐ **Network Security:** TCP/IP analysis, firewall rule management, and Wireshark proficiency.
- ☐ **Operating Systems:** Linux hardening, Windows Event Log analysis, and Active Directory hygiene.
- ☐ **Security Operations:** SIEM alert triage, EDR telemetry review, and incident escalation playbooks.

- ❑ **Cloud Security:** AWS/Azure IAM least-privilege configuration, VPC security, and CSPM posture management.
- ❑ **Governance & Compliance:** Risk assessment methodologies, NIST/ISO frameworks, and control testing.
- ❑ **Soft Skills & Ethics:** Executive communication, incident documentation, and professional ethical standards.

Final Conclusion: Cybersecurity is not a single skill—it is an integrated combination of technical knowledge, practical experience, problem-solving judgment, ethical rigor, and continuous learning. The objective is not to master every technology, but to develop the exact capabilities required for your role.

**UNDERSTAND THE SKILL. PRACTICE THE SKILL. DEMONSTRATE THE SKILL.
ASSESS THE SKILL. IMPROVE THE SKILL. BUILD SECURITY CAPABILITY.**

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